

MATHEMATICAL MANUSCRIPT ASSESSMENT

ReviewBot Scientific Referee Report

The Face-Adjacency Graph of the Truncated Octahedron

Laplacian Spectrum, Symmetry Decomposition, an Inter-Orbit Operator, and the Magnetic Laplacian

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Assessment: 8 October 2026

**Submission assessed.** The supplied 13-page Draft 6 is treated as a standalone note in finite graph spectra and discrete magnetic Laplacians. This review also examines the publicly available tagged source for the script changes expressly described in Section 8. Its scientific findings concern this note, not the author’s separate physical programme. Literature originality is assessed cautiously, not exhaustively.

|                           |                                                                                                                                           |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Scientific recommendation | Accept with minor revisions, subject to specialist-venue fit and final released-artifact checks                                           |
| Mathematical validity     | No confirmed critical or major mathematical error in the claims examined; a formal proof-assistant certification is not asserted          |
| Overall assessment        | Approximately 9.1/10 (rubric-weighted scientific judgment, not a correctness probability)                                                 |
| New findings              | 0 critical, 0 major; 2 minor release/editorial tasks and 2 optional clarifications                                                        |
| Draft 5 correction status | Five previously requested groups substantively addressed in Draft 6; independent reenactment of the complete tagged release not performed |

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## 1 Executive scientific assessment

**Recommendation.** Draft 6 is a materially improved mathematical note. It presents a coherent, checkable account of the combinatorial Laplacian of a non-regular 14-vertex face graph, explains its representation-theoretic multiplicities, and derives special magnetic spectra at uniform quarter and half flux. The author has incorporated the substantive reproducibility corrections requested after Draft 5, supplied a further spectral moment, and stated precisely what is and is not proof-assistant checked.

**Main strengths.** The ordinary spectrum is explained by small symmetry blocks rather than merely stated as a fourteen-dimensional determinant. The special quarter-flux spectrum has an explicit Gaussian-integer gauge, a verified Gram identity, invariant two-dimensional blocks, and a restricted intertwiner. The thirteen-spectrum theorem rests on a short analytic third-moment argument, not just plots. The five-parallelhedron comparison is supported by separate constructions and explicitly disclaims any physical cell-selection criterion.

**Remaining limitations.** The contribution is deliberately example-specific: only charges  $q = 0, 6, 12$  (and reflected charges) have exact eigenvalue lists, while the other sectors have an exact distinction theorem and a numerical sweep. Exhaustive historical priority is not established. An archived, tagged release is described and its public source was inspected, but the author's complete Draft 6 runner has not been freshly executed for this assessment; this is a separate, bounded reproducibility qualification, not evidence of an error in the spectra.

**Judgment of progress from Draft 5.** The five identified discrepancies in tolerances, unimodularity description, flux-lift ordering, review attribution, and the dependency declaration are addressed in the revised text and, where source was accessible, in the tagged programs. The author also incorporated the new fourth-moment identity and the exact tree count into Remark 6.3, and expanded the proof-status table. These changes improve precision and the reader's ability to verify the mathematics.

## 2 Category-level scoring

The paper is classified as pure mathematics with computational corroboration. The scoring uses 14 core criteria, with the applicable gauge and algebraic criteria; conceptual-breakthrough and interdisciplinary-unification scores are not activated. Core category masses (0.50, 0.25, 0.20, 0.05) are scaled to 90% after reserving 7% for gauge structure and 3% for algebraic structure. Within-category metric proportions are 10:12:10:8, 10:8:7, 7:7:6 and 5:3:4:3, respectively.

| Category                         | Weight      | Score / 10 | Contribution |
|----------------------------------|-------------|------------|--------------|
| Mathematical correctness (M1–M4) | 45.0%       | 9.25       | 4.163        |
| Completeness (M5–M7)             | 22.5%       | 9.26       | 2.083        |
| Clarity (M8–M10)                 | 18.0%       | 9.24       | 1.662        |
| Significance (M11–M14)           | 4.5%        | 6.22       | 0.280        |
| Gauge structure (M15)            | 7.0%        | 9.30       | 0.651        |
| Algebraic structure (M18)        | 3.0%        | 9.50       | 0.285        |
| <b>Total</b>                     | <b>100%</b> |            | <b>9.123</b> |

The reported rounded value is 9.1/10. These numbers summarize judgments about the *manuscript*; a verified mathematical defect would take precedence over any attractive score. The significance score is intentionally more restrained than the correctness score because a focused exact example is not a new general graph-theoretic framework.

### 3 Complete metric assessment

| Metric | Criterion               | Score | Manuscript-specific evidence and assessment                                                                                                                                                                                                                 |
|--------|-------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M1     | Theorem precision       | 9.2   | Propositions 2.1, 6.1 and 6.2 and Theorems 3.1, 5.1 and 6.4 identify the finite graph, flux conventions, sectors and operator domains. The intertwiner is explicitly restricted rather than extended to all of $\mathbb{C}^{14}$ .                          |
| M2     | Proof completeness      | 9.2   | The orbit decomposition, cube Fourier basis, Gram reduction, tree gauge and third-moment proof form coherent proof chains. Standard character identifications are compressed; the note supplies the needed dimensions and generators.                       |
| M3     | Calculation accuracy    | 9.4   | Reconstruction gives 14 vertices, 36 edges, 24 triangles, 42 four-cycles; the claimed ordinary, quarter and half characteristic polynomials agree. The fourth moment and spanning-tree count were independently reconstructed exactly.                      |
| M4     | Logical consistency     | 9.2   | The $A_{1g}$ line in the eigenvalue-7 space is correctly distinguished from $T_{2g}$ . The two scales $K^2 = -4I$ and $-12I$ are not confused. The $q = 6$ rotation representation uses the right symmetry labels.                                          |
| M5     | Definition coverage     | 9.2   | The face graph, $A, D, \Delta$ , orbit projectors and incidence blocks, support weight, gauge transformation and uniform sectors are defined. A brief one-line summary of the three separate exactness regimes would help newcomers.                        |
| M6     | Assumption transparency | 9.2   | The spherical embedding is stated for gauge uniqueness, and cube flatness is confined to $q = 6$ . Section 8 now distinguishes exact arithmetic from numerical tolerances and explicitly declares absence of proof-assistant checking.                      |
| M7     | Scope specification     | 9.4   | The manuscript excludes wider physical claims, arbitrary charge eigenvalue formulas, analytic interpolation of plotted lines and a canonical selection principle among space-filling cells. These restrictions are appropriate and consistently maintained. |
| M8     | Notation consistency    | 9.2   | The previously ambiguous integer lift is now $(-23, 1, \dots, 1)$ , explicitly tied to plaquette 0. The zero-flux $g/u$ labels are not wrongly transported into the magnetic representation.                                                                |
| M9     | Proof accessibility     | 9.3   | The quotient blocks, Hamming-distance Gram proof, five-step roadmap at quarter flux, and explicit $2 \times 2$ blocks make the results unusually teachable for a specialist note.                                                                           |
| M10    | Organization            | 9.2   | Logical sequence from graph geometry to ordinary spectrum, $K$ , magnetic flux, Fedorov comparison and reproducibility. The extensive acknowledgements compete slightly with the mathematical story.                                                        |
| M11    | Originality             | 6.0   | The explicit calculation is useful and appears nontrivial; the graph and underlying magnetic frameworks are established. No exhaustive priority investigation or general novelty theorem is supplied.                                                       |
| M12    | Generality              | 4.0   | Main closed-form analysis centres on one graph and the three solvable charges. The five-type table broadens the example collection but does not establish a universal selection or spectral theorem.                                                        |
| M13    | Literature context      | 8.7   | Relevant graph spectral, spherical magnetic, gain graph and Fedorov sources are cited; the author avoids asserting unverified priority.                                                                                                                     |
| M14    | Potential impact        | 5.5   | Provides an exact reference problem, a reusable illustration of symmetry decomposition and gauge compensation, and reproducibility exercises. Wider impact remains prospective.                                                                             |

| Metric | Criterion                       | Score | Manuscript-specific evidence and assessment                                                                                                                                                                |
|--------|---------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M15    | Gauge structure                 | 9.3   | The vertex gauge law, spherical face-holonomy classification, flat-cube gauge, charge conjugation and compensated proper-rotation action fit together without mistaking a chosen gauge for intrinsic data. |
| M16    | Claimed conceptual breakthrough | N/A   | Not activated: no broad paradigm-change claim is made within this note.                                                                                                                                    |
| M17    | Interdisciplinary synthesis     | N/A   | Not activated: speculative physical programme is expressly excluded from this mathematical submission.                                                                                                     |
| M18    | Algebraic structure             | 9.5   | Exact arithmetic, fields $\mathbb{Q}(\sqrt{17})$ and $\mathbb{Q}(\sqrt{17}, \sqrt{57})$ , representation blocks, Gram identities and the magnetic group law are coherent.                                  |

## 4 Detailed mathematical verification

### 4.1 Graph counts and the two ordinary characteristic polynomials (pp. 2–4)

Write  $S = \{(a, \sigma) : a = 1, 2, 3, \sigma = \pm 1\}$  for the six square faces and  $H = \{\pm 1\}^3$  for the eight hexagons. Two hexagons are adjacent when their signs differ in one coordinate; a square  $(a, \sigma)$  meets precisely the four  $h \in H$  with  $h_a = \sigma$ . Consequently, there are no square–square edges, 12 cube edges and 24 square–hexagon edges, hence 36 altogether, with degree sequence  $4^6 6^8$ .

There are 24 triangular faces in this dual embedding. For the adjacency matrix  $A$ , the closed-walk identity

$$\text{tr } A^4 = 8c_4 + 2 \sum_i d_i^2 - \sum_i d_i$$

gives  $c_4 = 42$  from  $\text{tr } A^4 = 1032$ ,  $\sum d_i^2 = 384$  and  $\sum d_i = 72$ . These counts were independently reconstructed from the stated adjacency rules.

The exact ordinary spectral factorizations are

$$\det(xI - A) = x^2(x+1)^3(x+3)(x^2-3x-12)(x^2-x-4)^3, \quad (1)$$

$$\det(xI - \Delta) = x(x-9)(x-7)^4(x-4)^2(x^2-9x+16)^3. \quad (2)$$

Both agree with independent exact matrix calculations.

### 4.2 Why the symmetry decomposition explains the spectrum (Theorem 3.1)

The two  $O_h$ -orbits carry

$$\mathbb{R}^6 \cong A_{1g} \oplus E_g \oplus T_{1u}, \quad \mathbb{R}^8 \cong A_{1g} \oplus T_{1u} \oplus T_{2g} \oplus A_{2u}.$$

The one-copy sectors immediately give eigenvalues 4 (multiplicity 2), 7 (multiplicity 3), and 9 (multiplicity 1). On orbit constants, the raw  $A_{1g}$  block is

$$\begin{pmatrix} 4 & -4 \\ -3 & 3 \end{pmatrix}, \quad \chi(x) = x(x-7),$$

while each of the three  $T_{1u}$  copies has raw block

$$\begin{pmatrix} 4 & -4 \\ -1 & 5 \end{pmatrix}, \quad \chi(x) = x^2 - 9x + 16.$$

Thus the two irrational eigenvalues are  $(9 \pm \sqrt{17})/2$  with multiplicity three. All 14 dimensions are accounted for. The multiplicity-four eigenvalue 7 is *not* a four-dimensional irreducible representation: it contains  $A_{1g} \oplus T_{2g}$ .

### 4.3 Orbit support and the antisymmetric operator (pp. 4–5)

The support weights in Proposition 4.1 agree with the projectors and the normalized  $T_{1u}$  eigenvectors. In particular

$$\omega_s(E_{r_1}) = \frac{1}{2}(1 + 1/\sqrt{17}), \quad \omega_s(E_{r_2}) = \frac{1}{2}(1 - 1/\sqrt{17}).$$

For  $K = P_s \Delta P_h - P_h \Delta P_s$ , the operator sends the raw orbit vectors  $s \mapsto h$ ,  $h \mapsto -4s$  in each  $T_{1u}$  plane, hence

$$K|_{T_{1u}} \sim \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}, \quad K^2 = -4I.$$

On the orbit-constant plane, instead,  $K\mathbf{1}_s = 3\mathbf{1}_h$  and  $K\mathbf{1}_h = -4\mathbf{1}_s$ , giving  $K^2 = -12I$ . This distinction matters: stating  $K^2 = -4I$  on the whole space or claiming that  $K$  kills the entire  $E_7$  would be wrong; the manuscript correctly avoids both.

### 4.4 Thirteen distinct magnetic spectra (Propositions 6.1–6.2)

The dual graph is cellularly embedded in a sphere with 14 vertices, 36 edges and 24 triangular faces. Its cycle rank is  $36 - 14 + 1 = 23$ , matching the single relation among the 24 oriented plaquette boundaries. Prescribed face holonomies with product one are therefore realizable, and a fixed set of face holonomies determines a gauge class. Uniform charges are  $q \bmod 24$ .

Writing  $\Delta_q = D - A_q$ , cyclicity of the trace gives

$$\text{tr } \Delta_q^3 = \text{tr } D^3 + 3 \text{tr } (DA_q^2) - \text{tr } A_q^3.$$

Here the first two terms are  $2112 + 3 \cdot 384 = 3264$ . The 24 oriented triangular plaquettes, each counted in three cyclic starting positions and both orientations, contribute

$$\text{tr } A_q^3 = 144 \cos(\pi q/12).$$

Therefore

$$\boxed{\text{tr } \Delta_q^3 = 3264 - 144 \cos(\pi q/12)}.$$

Equality of spectra forces equal third moments, hence  $p \equiv \pm q \pmod{24}$ ; complex conjugation supplies the converse. Representatives  $0, \dots, 12$  have distinct moments. The theorem establishes exactly 13 spectra without claiming explicit algebraic eigenvalues at every charge.

### 4.5 Quarter flux: a complete factorization with a hand gauge (pp. 7–9)

The matrix  $B$  of Eq. (7), with cube adjacency  $C$ , has exact Gram products

$$B^*B = (9I - C^2)/2 = 4(P_1 + P_2), \quad BB^* = 4I_6.$$

These identities were checked in Gaussian-integer arithmetic directly from the displayed formula. The cube eigenspaces  $W_1$  and  $W_2$  carry eigenvalues  $+1$  and  $-1$  (both dimensions 3). Each yields the invariant block

$$\begin{pmatrix} 4 & -2 \\ -2 & 6 - \mu \end{pmatrix} \quad (\mu = 1, -1),$$

with characteristic polynomials  $x^2 - 9x + 16$  and  $(x - 3)(x - 8)$ . The two remaining one-dimensional components give levels 3 and 9. Therefore

$$\det(xI - \Delta_6) = (x - 9)(x - 8)^3(x - 3)^4(x^2 - 9x + 16)^3.$$

The stated incidence intertwiner acts only on the six-dimensional sector, where the two incidence maps are isometries. The compensated symmetry matrices  $R_s(g) = BP(g)B^*/4$  were separately checked for all 24 proper signed-permutation rotations; they are monomial, unitary and satisfy the group law for all 576 ordered products. This is strong finite corroboration of the manuscript's hand proof.

#### 4.6 Half flux and the signless Laplacian (Theorem 6.4)

At  $q = 12$ , setting every oriented edge phase to  $-1$  realizes a  $-1$  holonomy around each triangular face. The magnetic Laplacian becomes  $D + A$ . Its  $A_{1g}$  block has characteristic polynomial  $x^2 - 13x + 24$ , the  $T_{1u}$  block has  $(x - 3)(x - 8)$ , and the remaining sectors yield the integer levels. Exact computation gives

$$\det(xI - \Delta_{12}) = (x - 8)^3(x - 5)^3(x - 4)^2(x - 3)^4(x^2 - 13x + 24),$$

matching the stated discriminant 73.

#### 4.7 New Draft 6 checks: the fourth moment and spanning trees (Remark 6.3)

A separately reconstructed oriented face–edge incidence matrix yields an integer uniform-flux lift and, by closed-walk enumeration, the following identities in the group ring of phases, with  $z^{24} = 1$ :

$$\begin{aligned}\mathrm{tr} \Delta(z)^3 &= 3264 - 72(z + z^{-1}), \\ \mathrm{tr} \Delta(z)^4 &= 25080 - 1536(z + z^{-1}) + 144(z^2 + z^{-2}) + 24(z^4 + z^{-4}).\end{aligned}$$

At  $z = \exp(2\pi i q/24)$  the second becomes

$$\mathrm{tr} \Delta_q^4 = 25080 - 3072 \cos \frac{\pi q}{12} + 288 \cos \frac{\pi q}{6} + 48 \cos \frac{\pi q}{3}.$$

It gives 22344 at  $q = 0$ , consistent with the eigenvalues from Theorem 3.1. A separate exact Laplacian cofactor computation gives

$$\tau(\Gamma) = 101\,154\,816,$$

matching the spectral Matrix–Tree expression  $9 \cdot 7^4 \cdot 4^2 \cdot 16^3/14$ . These are valid corroborating results, not independent evidence of a physical theory.

#### 4.8 Five Fedorov graph spectra (p. 10 and Appendix A)

Reconstructing the four comparison graphs from the adjacency descriptions— $K_{2,2,2}$ , a  $C_6$  joined to two nonadjacent apexes, the cube’s line graph and the stated twelve-vertex cyclic elongated-dodecahedron graph—gives the claimed characteristic polynomials:

$$\begin{aligned}P_{\text{cube}} &= x(x - 4)^3(x - 6)^2, \\ P_{\text{prism}} &= x(x - 3)^2(x - 5)^2(x - 6)^2(x - 8), \\ P_{\text{rhombic}} &= x(x - 2)^3(x - 4)^3(x - 6)^5, \\ P_{\text{elongated}} &= x(x - 2)(x - 4)^2(x - 6)^3(x - 8)(x^2 - 10x + 20)^2.\end{aligned}$$

The fifth row is the ordinary truncated-octahedron factorization above. Both nonintegral polynomial discriminants, 20 and 17, are correctly distinguished. The distinction is descriptive; it does not prove that Nature must select this cell.

## 5 Draft 5 correction closure and remaining issues

| ID | Previously identified problem                     | Draft 6 status                                                                                                                      |
|----|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| R1 | Numerical absolute tolerance contract             | <b>Addressed.</b> Section 8 declares <code>rtol = 0</code> and the tagged plotting source uses that setting in the compared checks. |
| R2 | Rounded floating determinant called “exact”       | <b>Addressed.</b> Integer Bareiss determinant calculation now appears in the tagged source; exact lift residual is checked.         |
| R3 | Ordered lift target disagreed with source         | <b>Addressed.</b> $(-23, 1, \dots, 1)$ with the compensating entry on plaquette 0 is stated consistently.                           |
| R4 | Overstrong descriptions of review independence    | <b>Substantially addressed.</b> Successive rounds are described as one sequence; attribution remains somewhat long.                 |
| R5 | <code>matplotlib</code> missing from requirements | <b>Addressed.</b> The tagged requirements file includes it and the manuscript gives installation instructions.                      |

The statuses above distinguish the *manuscript and inspected tagged source* from a fresh end-to-end execution of the author’s complete current release. The latter was not performed in this review. This limit does not undo the separate mathematical calculations summarized in Section 4.

### Minor task 1: final archival reproducibility (Section 8, pp. 10–11)

The manuscript promises a Zenodo archive of the submitted state but correctly leaves its identifier pending. At submission, provide the immutable archived DOI or exact full release commit, and ensure the cited scripts, CSV, figure, certificate and reviewer verification suites are present. Where possible, provide the final runner’s result for the archived version rather than relying on a mutable tag alone. The public plotting script header still refers to “Draft 5” although it is revision 2 for Draft 6; update that descriptive header to prevent confusion. **Status: minor release/editing item, not a demonstrated spectral defect.**

### Minor task 2: condense verification provenance (pp. 11–12)

The five historical review rounds now have more careful attribution, but several detailed paragraphs of review and software history take space from the mathematical narrative. Consider consolidating their scientific contributions and moving per-round execution history into a referenced record. Where an outside review method is named, confirm the parties authorize that description and avoid implying endorsement or independent human refereeing beyond the documented record. This is an editorial suggestion about precision and readability, not an allegation of misattribution.

### Optional clarity improvements

1. In the mathematical introduction or Section 8, offer a compact distinction between *hand-proved*, *exact finite computational corroboration*, *numerically checked* and *proof-assistant validated*. The current proof-status Table 2 already makes the essential distinction and may suffice with a shorter cross-reference.
2. The spectral curve caption correctly says the joining segments are not analytic eigenbranches. Repeating that qualifier near the first introduction of Figure 1 would help readers who skip detailed captions.

## 6 Effect of the remaining issues on scientific conclusions

| Claim group                                       | Verified support                                | Dependence on remaining item                                                |
|---------------------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------|
| Ordinary graph and spectrum (Prop. 2.1; Thm. 3.1) | Exact independent matrix arithmetic             | Neither minor task changes the stated factorization                         |
| Orbit weights and $K$ (Prop. 4.1; Thm. 5.1)       | Hand reductions; exact sector checks            | No mathematical dependency                                                  |
| Spherical flux and 13 spectra (Props. 6.1–6.2)    | Topological proof; exact walks                  | Release provenance affects reproducibility, not the logic                   |
| Quarter and half spectra (Thm. 6.4; Prop. 6.5)    | Exact Gaussian-integer blocks and signless form | Releasing archived scripts strengthens independent replay                   |
| Fourth moment and tree count (Remark 6.3)         | Exact enumerations/cofactor                     | Stable artifact citation helpful, not essential to the displayed identities |
| Fedorov comparison (Prop. 7.1; Appendix A)        | Independent exact graph constructions           | Neither minor task changes classification                                   |

No confirmed new mathematical defect propagates into the spectral conclusions. The remaining release issues affect archival traceability and presentation rather than the finite algebra.

## 7 Specialist scientific assessment

**Gauge.** The paper states the vertex-gauge law explicitly; preserves the gauge-invariant distinction between face holonomies and edge representatives; proves the spherical uniqueness claim under the correct topological assumption; and separates ordinary proper rotations at quarter flux from antiunitary/improper operations. The zero-sum real lift is not falsely equated with zero local flux.

**Algebraic structure.** The exact block decompositions resolve the square-root discriminants. The closed-walk formulas are independent polynomial invariants of the eigenvalue lists. The monomial magnetic action is a genuine representation, not merely a projective action asserted without a multiplier calculation.

**Formal proof status.** Table 2 openly reports that no Lean or other proof-assistant checking was performed. This is appropriate transparency. Exact finite arithmetic gives powerful reproducible support for its mapped finite claims; it should not be relabelled as full theorem proving by a proof assistant. The mathematical proofs themselves must stand on their stated arguments, as assessed above.

## 8 Recommendations to authors and editors

**For the authors:** retain the revised mathematical arguments, the new invariants and the proof-status table. Before final submission, ensure the archive is immutable and complete, correct the remaining “Draft 5” plotting-script header, and shorten provenance passages where they distract from the scientific contribution. Preserve the candid statement that the paper is a standalone finite-graph note and not a derivation of a preferred microscopic physical geometry.

**For the editor:** the manuscript appears suitable for a focused specialist venue in spectral graph theory, discrete magnetic operators or mathematical expository computation. No confirmed central mathematical error emerged from the examined proofs and finite checks. An author-side reproduction of the final archived release is a sensible last condition, not a substitute for the mathematical reasoning presented in the paper. A referee prioritizing broad new theory over exact worked examples may assign a different significance judgment; that is venue fit rather than correctness.

**Final recommendation: accept with minor revisions.** Draft 6 is a clearly improved and mathematically substantiated successor to Draft 5. This assessment is limited to the stated finite graph and its mathematical claims; it does not certify historical priority, complete software-release execution, general physical interpretations or machine-checked formal proof.